

AccountsManagerLib Unit Testing using NUnit

Introduction

This lab explains how to write automated unit test cases for the AccountsManagerLib project using the NUnit framework. The Login() functionality is tested for valid credentials, invalid credentials, and empty input values.

Objectives

- Learn automated unit testing using NUnit.
- Understand NUnit attributes such as TestFixture, Test and TestCase.
- Verify login functionality using different test scenarios.
- Use Assert.That() and follow the Single Assertion Rule.

Implementation

Step 1: Create a Class Library project named AccountsManagerLib.Tests in the same solution.

Step 2: Rename the default test class to AccountsManagerTests.cs.

Step 3: Add the reference of AccountsManagerLib to the test project.

Step 4: Install NUnit and NUnit3TestAdapter using NuGet Package Manager.

Step 5: Create test methods using the naming convention UnitUnderTest_Scenario_ExpectedOutcome.

Step 6: Write a test case for valid user_11 credentials and verify the welcome message.

Step 7: Write another test case for valid user_22 credentials and verify the welcome message.

Step 8: Write a test case for invalid credentials and verify the message 'Invalid user id/password'.

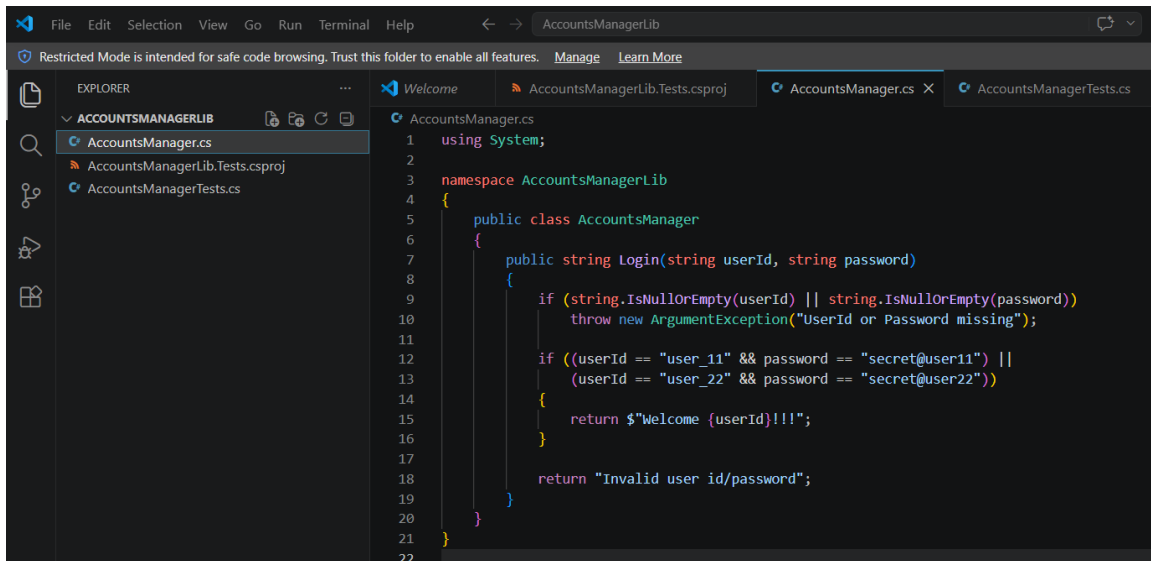
Step 9: Write a test case for empty user id or password and verify that ArgumentException is thrown.

Step 10: Run all the tests and verify that every test passes.

Step 11: Modify the Login() method intentionally and rerun the tests to observe failed test cases.

Execution

Paste the screenshot of the successful NUnit Test Explorer execution or terminal output here.



The screenshot shows the Visual Studio IDE with the AccountsManagerLib project open. The Explorer pane on the left shows the project structure with the following files:

- ACCOUNTSMANAGERLIB
 - AccountsManager.cs
 - AccountsManagerLib.Tests.csproj
 - AccountsManagerTests.cs

The main editor displays the AccountsManager.cs file with the following code:

```
1 using System;
2
3 namespace AccountsManagerLib
4 {
5     public class AccountsManager
6     {
7         public string Login(string userId, string password)
8         {
9             if (string.IsNullOrEmpty(userId) || string.IsNullOrEmpty(password))
10                 throw new ArgumentException("UserId or Password missing");
11
12             if ((userId == "user_11" && password == "secret@user11") ||
13                 (userId == "user_22" && password == "secret@user22"))
14             {
15                 return $"Welcome {userId}!!!";
16             }
17
18             return "Invalid user id/password";
19         }
20     }
21 }
22
```

Expected Result

The application should display the correct welcome message for valid credentials, display 'Invalid user id/password' for incorrect credentials, and throw an ArgumentException when either the user ID or password is empty.

Conclusion

This experiment helped in understanding NUnit unit testing, assertions, exception handling, and validating different login scenarios. The test cases confirmed that the login functionality behaves correctly under different conditions.